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(54) **LIGHT UNIFORM MODULE AND PROJECTION DEVICE**

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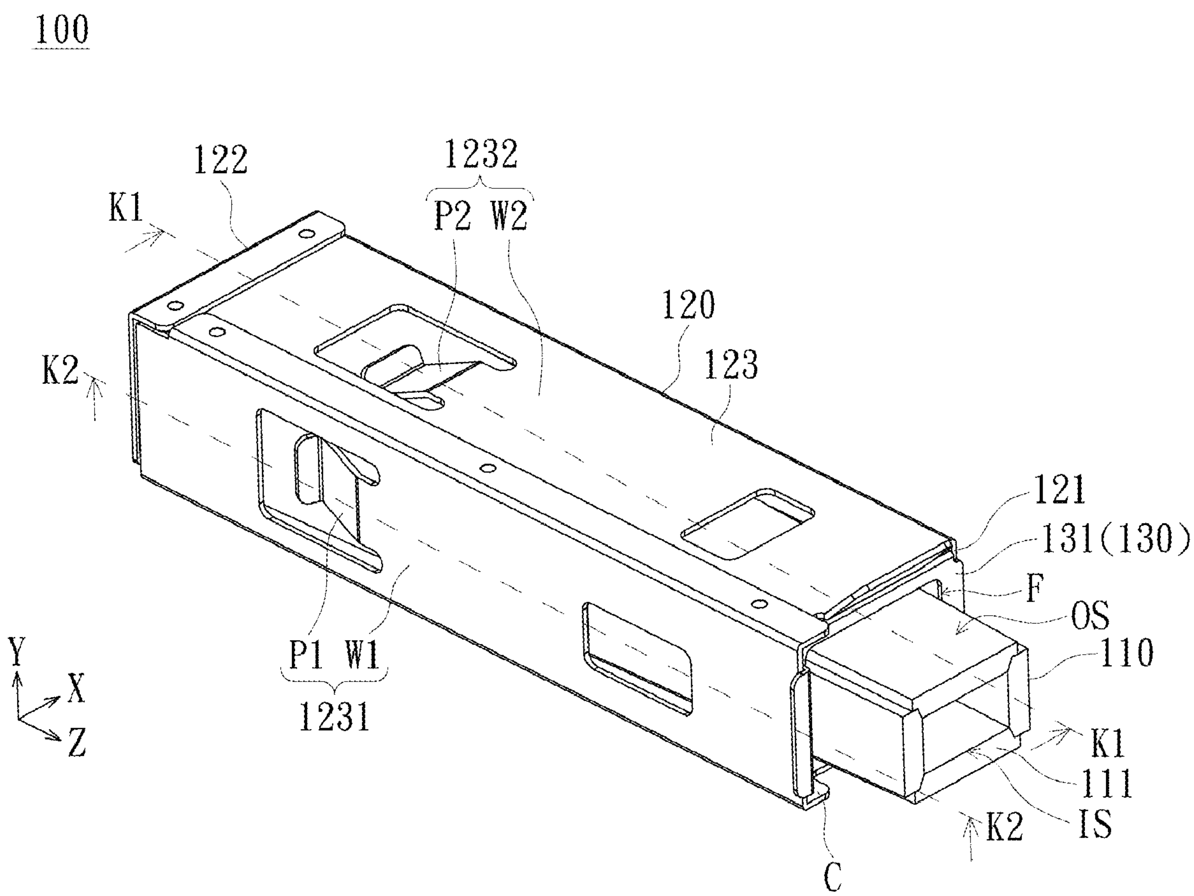
(57) **ABSTRACT**

A light uniform module includes a light uniform column, a sleeve, and an end fastening member. The light uniform column has a light inlet end and a light outlet end. The light inlet end has a light inlet surface. The light outlet end has a light outlet surface. The sleeve has an open end and a stop end. The sleeve is sleeved outside the light uniform column, and the sleeve accommodates the light uniform column. The end fastening member is disposed at the open end of the sleeve, and the end fastening member fastens relative positions of the light uniform column and the sleeve. The end fastening member has a frame to form a clamping opening, and the light inlet end of the light uniform column penetrates the clamping opening and stretches out of the sleeve from the open end of the sleeve. A projection device is also provided.



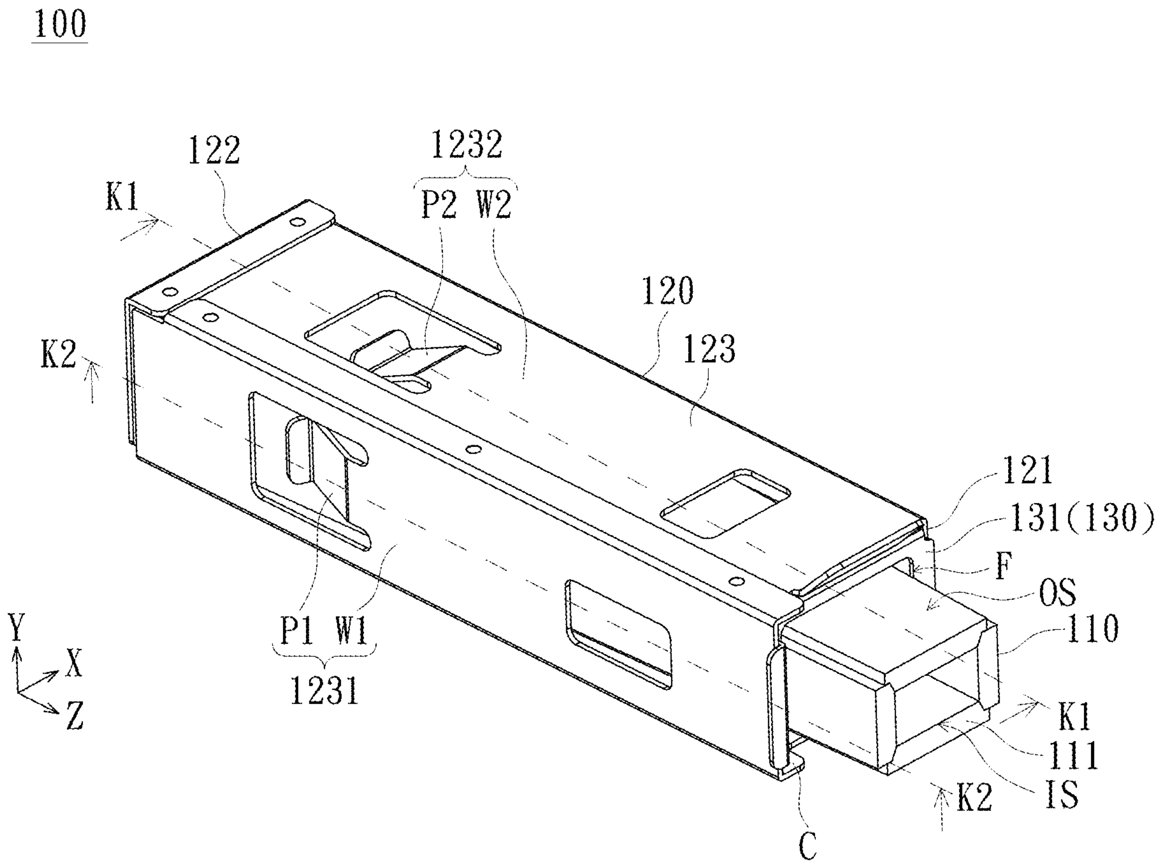
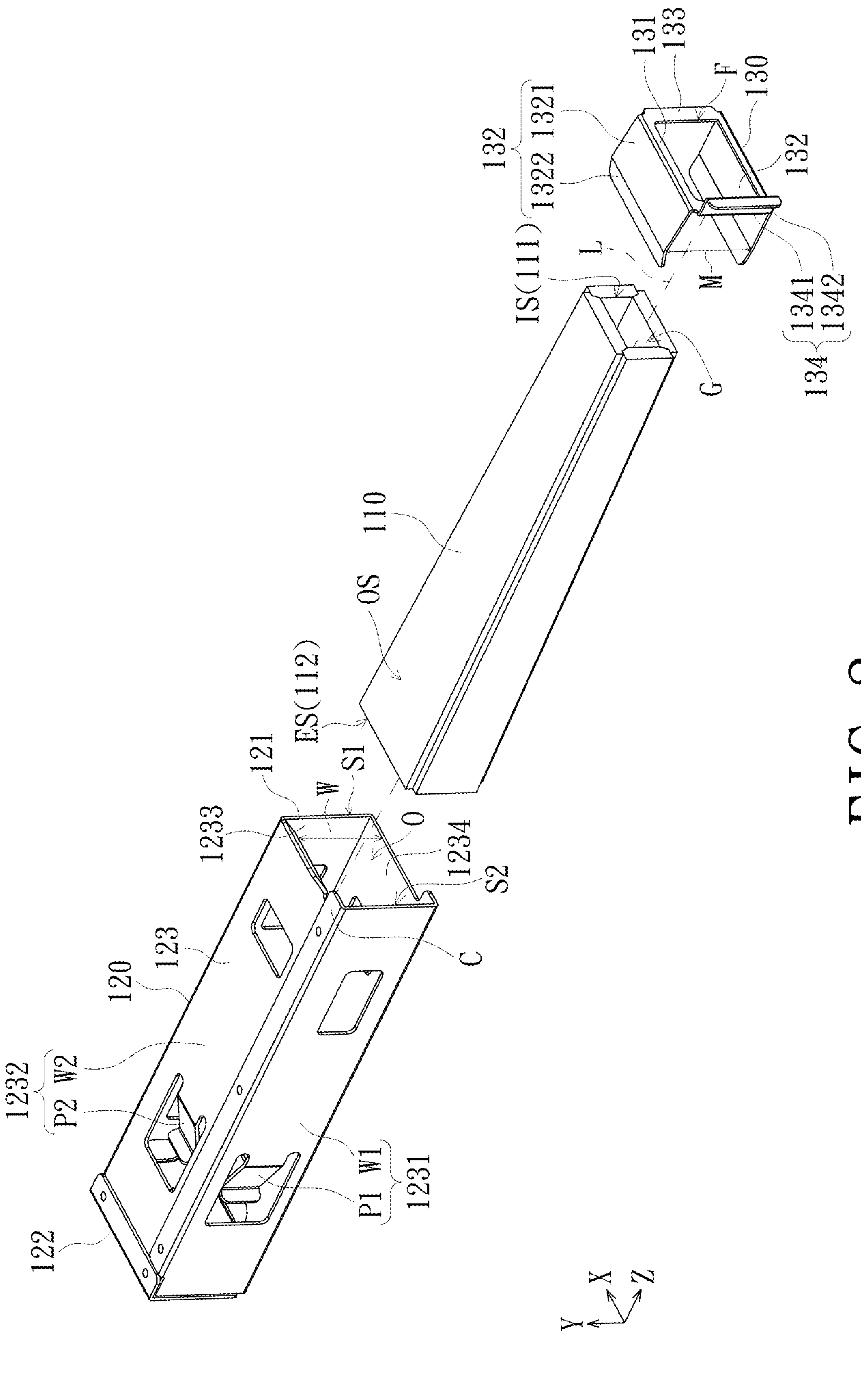

100
122
1232
P2 W2
K1
120
K2
123
121
131(130)
F
OS
110
Y
X
Z
P1 W1
1231
K1
111
IS
K2
C

FIG. 1


131
133
F
130
132
1321
1322
132
L
IS(111)
M
1341
1342
134
G
110
OS
ES(112)
121
W
S1
1233
O
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122
X
Y
Z

FIG. 2

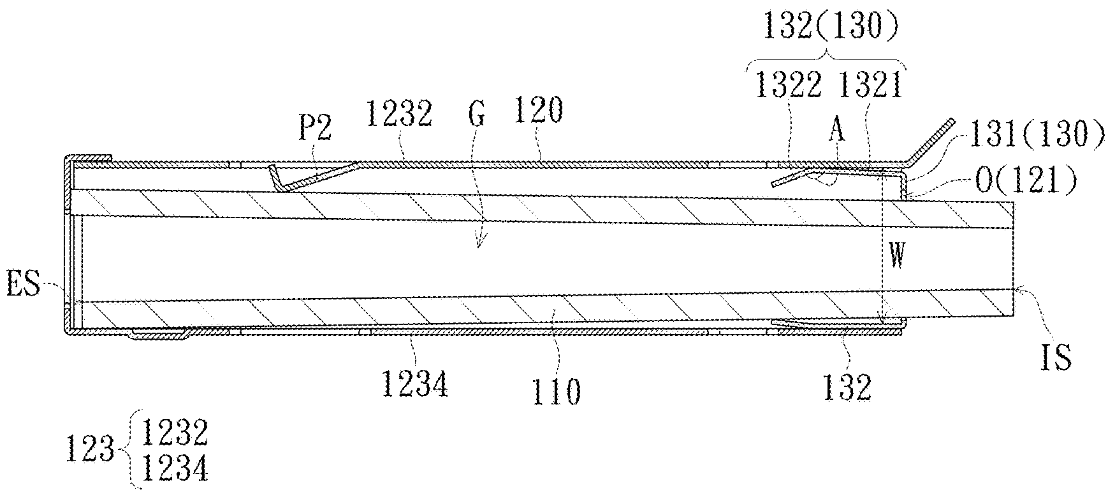

FIG. 3

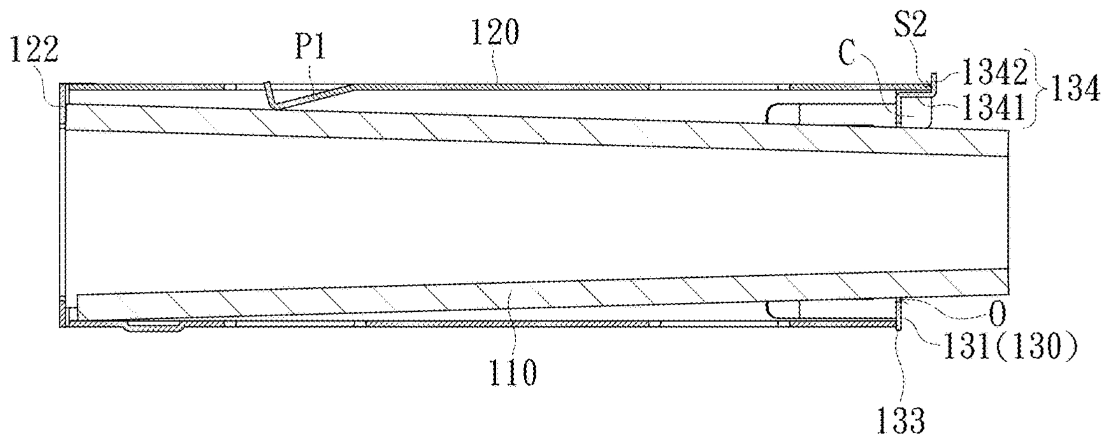

FIG. 4

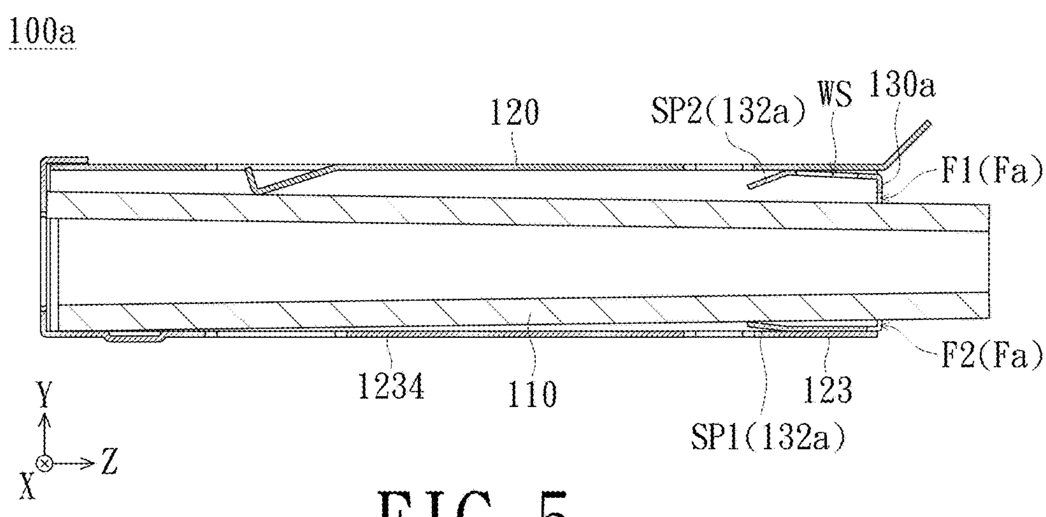

FIG. 5

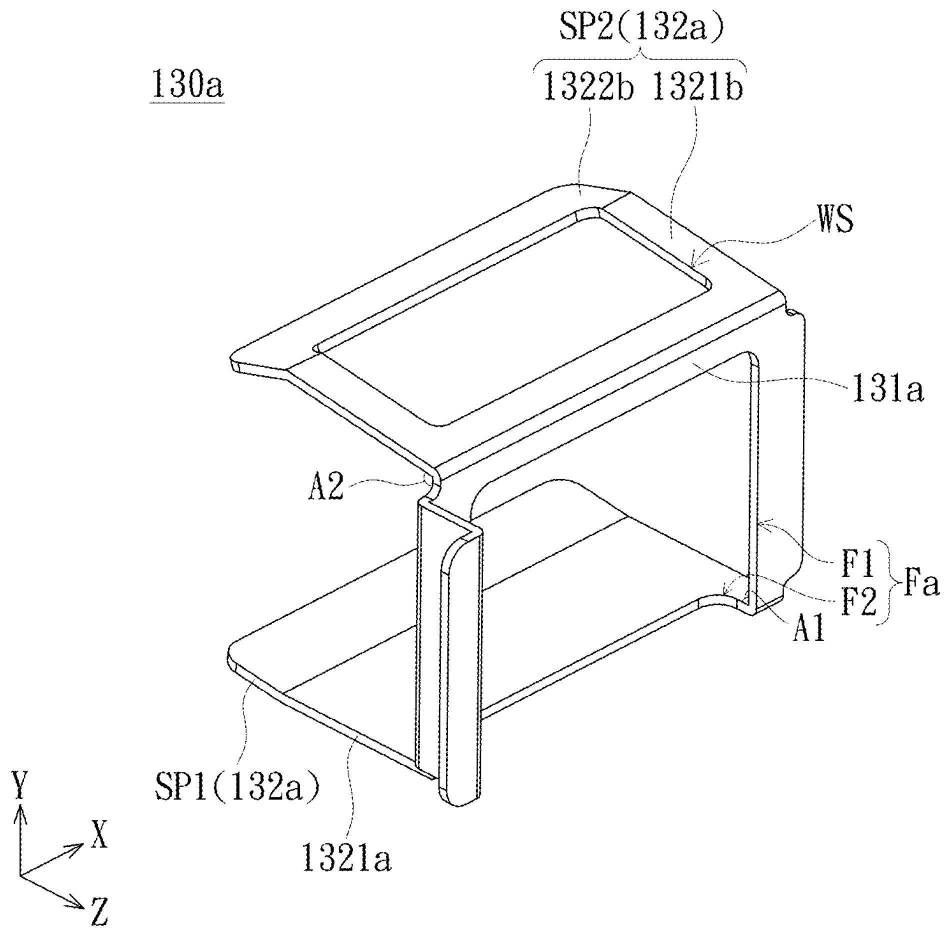
SP2(132a)
130a
1322b 1321b
WS
131a
A2
F1
Fa
F2
A1
Y
X
Z
SP1(132a)
1321a

FIG. 6

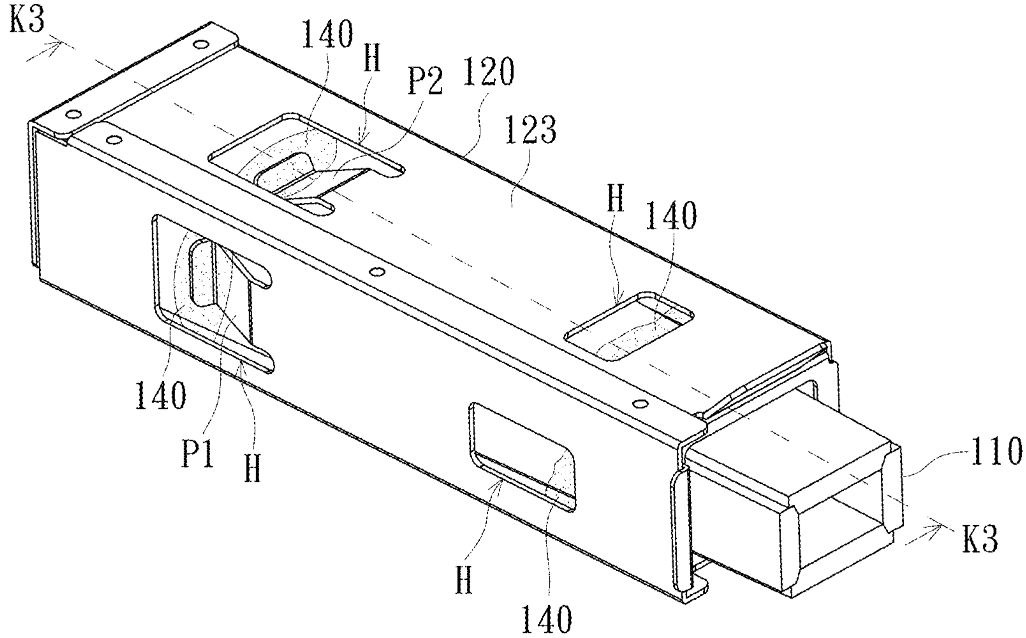
K3
140
H
P2
120
123
H
140
140
P1
H
110
K3
H
140

FIG. 7

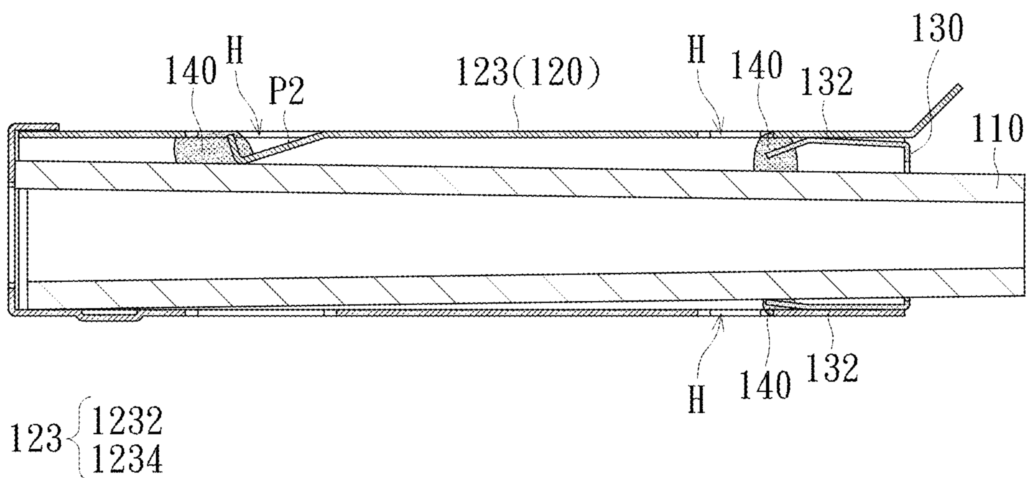
140
H
P2
123(120)
H
140
132
130
110
H
140
132
123
1232
1234

FIG. 8

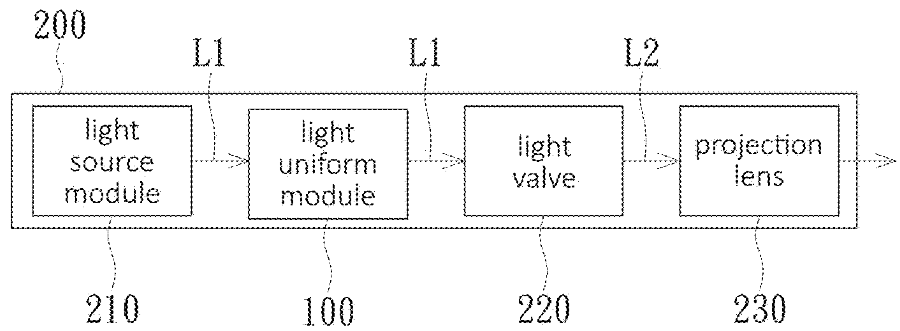
200
L1
L1
L2
light source module
light uniform module
light valve
projection lens
210
100
220
230

FIG. 9

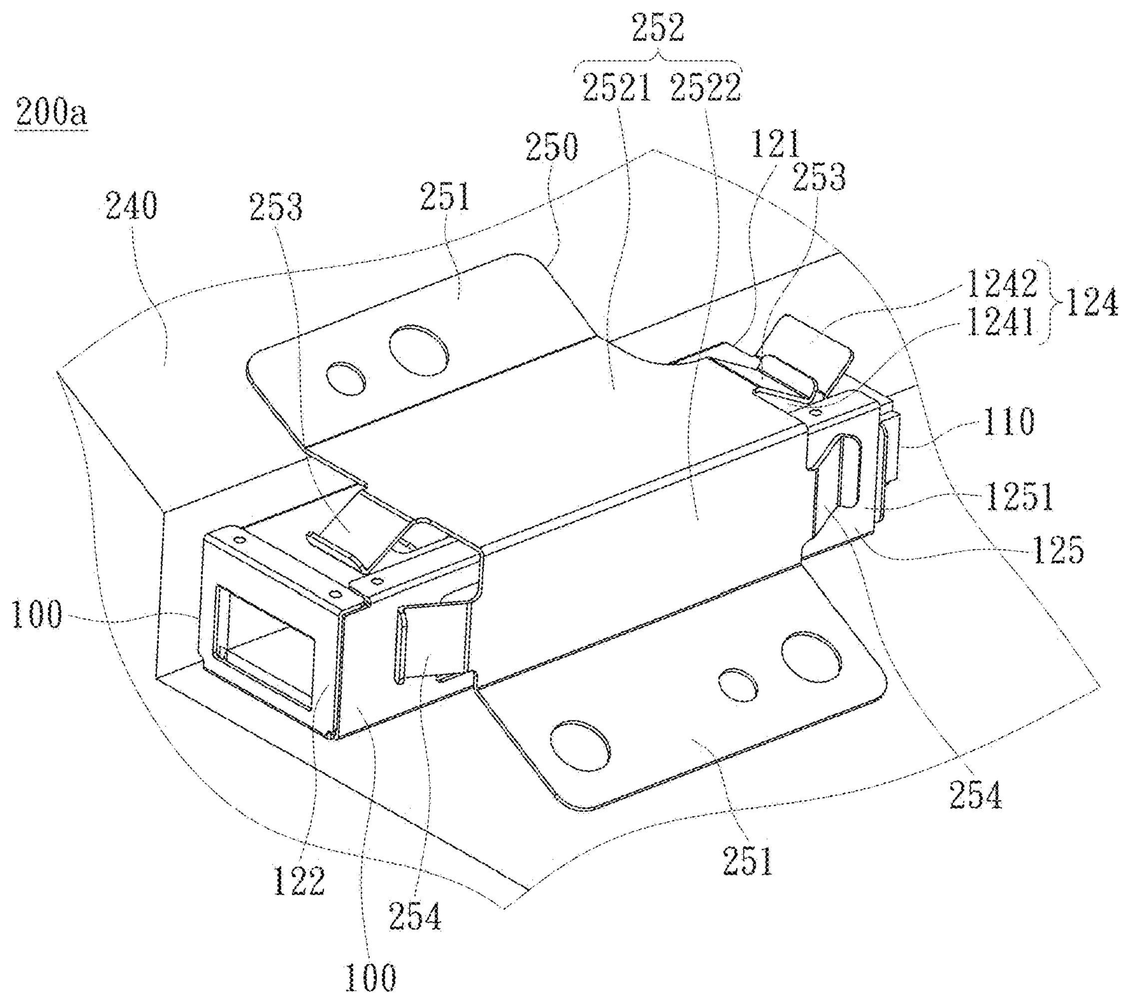
252
2521
2522
200a
250
121
253
240
253
251
1242
124
1241
110
1251
125
100
254
251
122
254
100

FIG. 10

LIGHT UNIFORM MODULE AND PROJECTION DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims the priority benefit of the China application (No. 202420990210.9) filed on May 9, 2024 and the U.S. provisional patent application (No. 63/563,929) filed on Mar. 11, 2024. The entirety of the above-mentioned patent applications is hereby incorporated by reference herein and made a part of this specification.

TECHNICAL FIELD

[0002] The disclosure relates to an optical module, and in particular to a light uniform module and a projection device provided with the light uniform module.

BACKGROUND

[0003] Types of light source used in projection devices have evolved from ultra-high pressure mercury lamp (UHP lamp) and light-emitting diode (LED) to laser diode (LD) light source with the market's requirements for brightness, color saturation, service life, non-toxic and environmental protection, and the like of the projection devices. In addition, the projection device is also provided with a light uniform module to homogenize beams generated by a light source module and transmit homogenized beams to a subsequent optical component.

[0004] Generally, a light uniform column and a mechanism are provided for the light uniform module, and the light uniform column is fastened in the projection device by using the mechanism. However, to fasten the light uniform column accurately, the existing mechanism needs to be pressed against the light uniform column tightly, resulting in excessive interference of the mechanism with the light uniform column. In this way, the light uniform module is not easily assembled, and the light uniform column is easily scratched by the mechanism. On the contrary, if the interference amount of the mechanism with the light uniform column is reduced, the light uniform column may loosen easily.

[0005] The information disclosed in this "BACKGROUND" section is only for enhancement understanding of the background and therefore it may contain information that does not form the prior art that is already known to a person of ordinary skill in the art. Furthermore, the information disclosed in this "BACKGROUND" section does not mean that one or more problems to be solved by one or more embodiments of the disclosure were acknowledged by a person of ordinary skill in the art.

SUMMARY

[0006] A light uniform module in an embodiment of the disclosure includes a light uniform column, a sleeve, and an end fastening member. The light uniform column is provided with a light inlet end and a light outlet end opposite to each other. The light inlet end is provided with a light inlet surface. The light outlet end is provided with a light outlet surface, and an area of the light inlet surface is smaller than an area of the light outlet surface. The sleeve is provided with an open end and a stop end opposite to each other. The sleeve is sleeved outside the light uniform column, and the sleeve is configured to accommodate the light uniform column. The light outlet end of the light uniform column faces the stop end. The end fastening member is disposed at the open end of the sleeve, and the end fastening member is configured to fasten relative positions of the light uniform column and the sleeve. The end fastening member is provided with a frame to form a clamping opening, and the light inlet end of the light uniform column penetrates the clamping opening and stretches out of the sleeve from the open end of the sleeve.

[0007] A projection device in an embodiment of the disclosure includes a light source module, the light uniform module, a light valve, and a projection lens. The light source module is configured to provide an illumination beam. The light uniform module is disposed on a transmission path of the illumination beam. The light uniform module is configured to receive the illumination beam, and the illumination beam enters a light uniform column through a light inlet surface. The illumination beam is totally reflected in the light uniform column and transmitted to a light outlet surface and leaves the light uniform column from the light outlet surface. The light valve is disposed on the transmission path of the illumination beam and is configured to convert the illumination beam emitted from the light uniform module to an image beam. The projection lens is disposed on a transmission path of the image beam and is configured to project the image beam.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The accompanying drawings are included to provide a further understanding of the disclosure, and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments of the disclosure and, together with the description, serve to explain the principles of the disclosure.

[0009] FIG. **1** is a schematic diagram of a light uniform module according to an embodiment of the present disclosure;

[0010] FIG. **2** is a schematic exploded diagram of the light uniform module in FIG. **1**;

[0011] FIG. **3** is a schematic cross-sectional view of the light uniform module, taken along the line K**1**-K**1** in FIG. **1**;

[0012] FIG. **4** is a schematic cross-sectional view of the light uniform module, taken along the line K**2**-K**2** in FIG. **1**;

[0013] FIG. **5** is a schematic cross-sectional diagram of a light uniform module according to another embodiment of the present disclosure;

[0014] FIG. **6** is a schematic diagram of an end fastening member in FIG. **5**;

[0015] FIG. **7** is a schematic cross-sectional diagram of a light uniform module according to another embodiment of the present disclosure;

[0016] FIG. **8** is a schematic cross-sectional view of the light uniform module, taken along the line K**3**-K**3** in FIG. **7**;

[0017] FIG. **9** is a schematic block diagram of a projection device according to an embodiment of the present disclosure; and

[0018] FIG. **10** is a schematic diagram of a projection device according to another embodiment of the present disclosure.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0019] In the following detailed description of the preferred embodiments, reference is made to the accompanying drawings which form a part hereof, and in which is shown by way of illustration specific embodiments in which the disclosure may be practiced. In this regard, directional terminology, such as "top", "bottom", "front", "back", etc., is used with reference to the orientation of the Figure(s) being described. The components of the disclosure can be positioned in a number of different orientations. As such, the directional terminology is used for purposes of illustration and is in no way limiting. On the other hand, the drawings are only schematic and the sizes of components may be exaggerated for clarity. It is to be understood that other embodiments may be utilized and structural changes may be made without departing from the scope of the disclosure. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of "including", "comprising", or "having" and variations thereof herein is meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless limited otherwise, the terms "connected", "coupled", and "mounted" and variations thereof herein are used broadly and encompass direct and indirect connections, couplings, and mountings. Similarly, the terms "facing", "faces", and variations thereof herein are used broadly and encompass direct and indirect facing, and "adjacent to" and variations thereof herein are used broadly and encompass directly and indirectly "adjacent to". Therefore, the description of "A" component facing "B" component herein may contain the situations that "A" component facing "B" component directly or one or more additional components is between "A" component and "B" component. Also, the description of "A" component "adjacent to" "B" component herein may contain the situations that "A" component is directly "adjacent to" "B" component or one or more additional components is between "A" component and "B" component. Accordingly, the drawings and descriptions will be regarded as illustrative in nature and not as restrictive.

[0020] The disclosure provides a light uniform module, which is easily assembled, with an assembly yield and assembly firmness taken into account.

[0021] The disclosure provides a projection device, which can improve image quality, with the product yield and product durability taken into account.

[0022] Other objectives, features and advantages of the disclosure will be further understood from the further technological features disclosed by the embodiments of the disclosure wherein there are shown and described preferred embodiments of this disclosure, simply by way of illustration of modes best suited to carry out the disclosure.

[0023] FIG. **1** is a schematic diagram of a light uniform module according to an embodiment of the disclosure. FIG. **2** is a schematic exploded diagram of the light uniform module in FIG. **1**. FIG. **3** is a schematic cross-sectional view of the light uniform module, taken along the line **K1-K1** in FIG. **1**. FIG. **4** is a schematic cross-sectional view of the light uniform module, taken along the line **K2-K2** in FIG. **1**. Refer to FIG. **1** and FIG. **2**. A light uniform module **100** includes a light uniform column **110**, a sleeve **120**, and an end fastening member **130**. The light uniform column **110** is provided with a light inlet end **111** and a light outlet end **112** (drawn in FIG. **2**) opposite to each other. The light inlet end **111** is provided with a light inlet surface IS. The light outlet end **112** is provided with a light outlet surface ES (drawn in FIG. **2**), and an area of the light inlet surface IS is smaller than an area of the light outlet surface ES. The sleeve **120** is provided with an open end **121** and a stop end **122** opposite to each other. The sleeve **120** is sleeved outside the light uniform column **110**, and the sleeve **120** is configured to accommodate the light uniform column **110**. The light outlet end **112** of the light uniform column **110** faces the stop end **122**. The end fastening member **130** is disposed at the open end **121** of the sleeve **120**, and the end fastening member **130** is configured to fasten relative positions of the light uniform column **110** and the sleeve **120**. The end fastening member **130** is provided with a frame **131** to form a clamping opening F, and the light inlet end **111** of the light uniform column **110** penetrates the clamping opening F and stretches out of the sleeve **120** from the open end **121** of the sleeve **120**.

[0024] The light uniform column **110** may be configured to receive an illumination beam. Specifically, the illumination beam can enter the light uniform column **110** through the light inlet surface IS of the light inlet end **111**, is totally reflected in the light uniform column **110** and transmitted to the light outlet end **112**, and then leaves the light uniform column **110** from the light outlet surface ES. In detail, the light uniform column **110** in this embodiment may be tube-shaped, and therefore, the illumination beam may be transmitted in light guide space G of the light uniform column **110**. A sectional area of the light uniform column **110** perpendicular to an optical axis L (drawn in FIG. **2**) gradually increases from the light inlet end **111** to the light outlet end **112**, for example, the light uniform column **110** in this embodiment may be a tapered rod **110**. It should be noted that, because most of the illumination beams are transmitted in the light guide space G of the light uniform column **110**, the above sectional area may be a sectional area perpendicular to the optical axis L in the light guide space G.

[0025] The sleeve **120** may be provided with a structure for interfering with the light uniform column **110** so that one side of the light uniform column **110** close to the light outlet end **112** is fastened in the sleeve **120**. The sleeve **120**, for example, also has a sidewall **123**. The side wall **123** is located between the open end **121** and the stop end **122** and surrounds the light uniform column **110**. The side wall **123** has a wall part **1231** and a wall part **1232** adjacent to each other. The wall part **1231** has a wall body **W1** and a pressing piece **P1** connected to each other, the wall part **1232** has a wall body **W2** and a pressing piece **P2** connected to each other, and the pressing piece **P1** and the pressing piece **P2** are separately configured to press an outer surface OS (drawn in FIG. **2**) of the light uniform column **110** close to the light outlet end **112**. In this way, the pressing piece **P1** and the pressing piece **P2** can separately fasten the light uniform column **110** in two different directions. In detail, the side wall **123** may also have a wall part **1233** and a wall part **1234** (both marked in FIG. **2**), and the pressing piece **P1**, for example, pushes the light uniform column **110** against the wall part **1233** in a positive direction X, to prevent the light uniform column **110** from moving in the direction X. Similarly, the pressing piece **P2** can push the light uniform column **110** against the wall part **1234** in a negative direction Y to prevent the light uniform column **110** from moving in the direction Y. The wall part **1231** and the wall part **1232** are substantially perpendicular, but the disclosure is not limited thereto.

[0026] Incidentally, the wall body W**1** and the pressing piece P**1** in this embodiment are, for example, integrally formed, and the wall body W**2** and the pressing piece P**2** may also be integrally formed. Specifically, a part of the wall part **1231** may be cut and bent to form the pressing piece P**1**, an unbent part of the wall part **1231** may form the wall body W**1**, and a processing manner for the pressing piece P**2** and the wall body W**2** is roughly the same as a processing manner for the pressing piece P**1** and the wall body W**1**. In an embodiment, the pressing piece P**1** is, for example, a piece additionally connected to the wall body W**1**, and the pressing piece P**2** may be a piece additionally connected to the wall body W**2**. In other words, the pressing piece P**1** and the wall part W**1** may be split structures, and the pressing piece P**2** and the wall body W**2** may also be split structures. For example, the pressing piece P**1** may be connected to the wall body W**1** by sticking, locking, or the like, and the pressing piece P**2** may be connected to the wall body W**2** in a manner that is roughly the same as a manner in which the pressing piece P**1** is connected to the wall body W**1**. In this embodiment, a material of the sleeve **120** may include metal, and the metal may include stainless steel, but the disclosure is not limited thereto.

[0027] In this embodiment, the end fastening member **130** may abut between the light uniform column **110** and the sleeve **120** to interfere with one side of the light uniform column **110** close to the light inlet end **111**. For example, an opening area of the clamping opening F is between an area of the light inlet surface IS and an area of the light outlet surface ES of the light uniform column **110**, so that the light uniform column **110** is interfered with the frame **131** to be fastened in the clamping opening F. Further, the frame **131** may be fastened to the open end **121** of the sleeve **120**, and an outer surface OS of the light uniform column **110** is clamped through the clamping opening F, so that one side of the light uniform column **110** close to the light inlet end **111** is fastened to the open end **121**, and the light inlet end **111** of the light uniform column **110** stretches out of the sleeve **120** from the open end **121**.

[0028] Refer to FIG. **2** and FIG. **3**. In addition, the end fastening member **130** may further be provided with two elastic pieces **132**, and the two elastic pieces **132** are respectively connected to two opposite sides of the frame **131**. When the end fastening member **130** is fastened to the open end **121** of the sleeve **120**, the two elastic pieces **132** are located in the sleeve **120**, and the two elastic pieces **132** separately push against the side wall **123** in directions away from each other. In this way, the end fastening member **130** may be fastened in the sleeve **120** by interference of the two elastic pieces **132** with the side wall **123**. In detail, the open end **121** of the sleeve **120** may have an open part O, and a maximum spacing M (marked in FIG. **2**) between the two elastic pieces **132** may be greater than the width W of the open part O. Therefore, after extending into the sleeve **120** from the open part O, the two elastic pieces **132** may be compressed by the wall part **1232** and the wall part **1234** to produce a reaction force to push against the wall part **1232** and the wall part **1234**, and then fastened in the sleeve **120** by interfering with the side wall **123**. Incidentally, a material of the end fastening member **130** may include metal, and the metal includes, for example, stainless steel, but the disclosure is not limited thereto.

[0029] In this embodiment, each of the two elastic pieces **132** may be provided with a pushing part **1321** and a guide part **1322**. The pushing part **1321** is connected to the frame **131**, and the frame **131** and the guide part **1322** are located on two opposite sides of the pushing part **1321**. The guide part **1322** is connected to the pushing part **1321**, and the guide part **1322** is bent toward the other guide part **1322** relative to the connected pushing part **1321**. Therefore, the end fastening member **130** can more easily extend into the sleeve **120** through the guidance of the guide part **1322**, so that the light uniform module **100** is assembled more easily. Incidentally, an angle A (marked in FIG. **3**) between the guide part **1322** and the pushing part **1321** of each of the two elastic pieces **132** is, for example, an obtuse angle, to further enhance the guidance effect provided by the guide part **1322** when the end fastening member **130** extends into the sleeve **120**. It should be noted that the above maximum spacing M between the two elastic pieces **132** is, for example, a spacing between two junctions of the two pushing parts **1321** and the two second guide parts **1322** that are connected correspondingly.

[0030] Refer to FIG. **2** and FIG. **4**. The end fastening member **130** may further be provided with at least one limiting structure, and for example, one limiting structure **133** (drawn in FIG. **2**) and one limiting structure **134** are provided in this embodiment. The limiting structure **133** and the limiting structure **134** are connected to the frame **131**, and the limiting structure **133** is configured to lean against the open end **121** from a supporting surface S**1** (drawn in FIG. **2**) of the open end **121** of the sleeve **120**, and the limiting structure **134** is configured to lean against the open end **121** from a supporting surface S**2** (drawn in FIG. **2**), to control a distance for which the end fastening member **130** extends into the sleeve **120**. A shape of the limiting structure **133** and the limiting structure **134** may be changed based on a shape of the open end **121** of the sleeve **120**. For example, the limiting structure **133** may be located between the two elastic pieces **132** and can protrude from the frame **131** along the positive direction X, to lean against the supporting surface S**1**. Further, the open end **121** is, for example, frame-shaped, and an end surface on a frame-shaped side may be the supporting surface S**1**.

[0031] In one embodiment, the open end **121** of the sleeve **120** may further be provided with, for example, a protruding part C. The light uniform column **110** stretches out of the open part O. The protruding part C protrudes from the open part O in a positive direction Z away from the stop end **122**, and the supporting surface S**2** is located at the protruding part C. The limiting structure **134** is, for example, provided with an extension section **1341** and a stop section **1342**. The extension section **1341** is connected to the frame **131**, and the frame **131** and the stop section **1342** are located on two opposite sides of the extension section **1341**. The stop section **1342** is connected to the extension section **1341** and is bent relative to the extension section **1341**. The stop section **1342** is configured to lean against the protruding part C from the supporting surface S**2** of the sleeve **120**. Simply, the extension section **1341** may extend along the positive direction Z away from the open end **121** to match the length of the protruding part C protruding from the open part O, so that the stop section **1342** can lean against the protruding part C when the end fastening member **130** extends into the sleeve **120**, and then a distance for which the end fastening member **130** extends into the sleeve **120** can be controlled. Still refer to FIG. **2**. In this embodiment, the limiting structure **133** and the limiting structure **134** are opposite each other, and the stop section **1342** and the limiting structure **133** can protrude from the frame **131** in opposite directions. However, in one embodiment, the end fastening member **130** may be provided with only the limiting structure **133** or the limiting structure **134**. In another embodiment, a quantity of the limiting structure **133** or the limiting structure **134** may be one or more than two, and the disclosure is not limited thereto.

[0032] Still refer to FIG. **1** and FIG. **2**. The assembly steps for the light uniform module **100** in this embodiment are described below. First, the light outlet end **112** of the light uniform column **110** is extended into the sleeve **120** from the open part O of the open end **121** of the sleeve **120**, until the light uniform column **110** is fastened in the sleeve **120** by interference of the two pressing pieces P**1** and P**2** on one side close to the light outlet end **112**. In addition, the stop end **122** can stop the light outlet end **112** of the light uniform column **110** in the sleeve **120**, and the light inlet end **111** of the light uniform column **110** stretches out of the sleeve **120** from the open part O. Next, the end fastening member **130** is sleeved on the light uniform column **110** from the light inlet end **111** of the light uniform column **110**, until the end fastening member **130** is fastened between the light uniform column **110** and the sleeve **120** by the interference of the light uniform column **110** and the sleeve **120**. The assembly for the light uniform module **100** is completed after the end fastening member **130** is fastened between the light uniform column **110** and the sleeve **120**. Incidentally, the above assembly steps are only an example and are not intended to limit the disclosure.

[0033] Compared with the existing technologies, for the light uniform module **100** in this embodiment, the sleeve **120** is used to fasten one side of the light uniform column **110** close to the light outlet end **112**, and the end fastening member **130** is used to fasten the light uniform column **110** on one side close to the light inlet end **111**. In this way, a structure used to interfere with the light uniform column **110** near the open end **121** may be omitted for the sleeve **120**, so that a thicker side (i.e., the side close to the light outlet end **112**) of the light uniform column **110** is not damaged by excessive interference of the structure while extending into the sleeve **120** from the open end. In addition, the end fastening member **130** uses the clamping opening F of the frame **131** to interfere with a thinner side (i.e., the side close to the light inlet end **111**) of the light uniform column **110**. Therefore, when the end fastening member **130** is sleeved from the light inlet end **111** to the light uniform column **110**, the frame **131** does not touch the light uniform column **110** before interfering with the light uniform column **110**, and the frame **131** can achieve the effect of fastening the light uniform column **110** by slight interference, thereby reducing the damage of the end fastening member **130** to the light uniform column **110**. In addition, because the end fastening member **130** and the sleeve **120** both have a function of fastening the light uniform column **110**, an interference amount of the sleeve **120** to the light uniform column **110** can be further reduced. In this case, the light uniform column **110**, the sleeve **120**, and the end fastening member **130** can still be firmly fastened to each other. Based on the above structure, the light uniform module **100** in this embodiment has an advantage of being easily assembled, with an assembly yield and assembly firmness taken into account.

[0034] FIG. **5** is a schematic cross-sectional diagram of a light uniform module according to another embodiment of the disclosure. FIG. **6** is a schematic diagram of an end fastening member in FIG. **5**. A structure and advantages of the light uniform module **100***a* in this embodiment are similar to the structure and advantages of the embodiment in FIG. **1**, and only differences are described below. Refer to FIG. **5** and FIG. **6**. Two elastic pieces **132***a* of the end fastening member **130***a* include a first elastic piece SP**1** and a second elastic piece SP**2**. A clamping opening Fa may have a clamping region F**1** and an extension region F**2** communicated to each other. A light uniform column **110** penetrates the clamping region F**1**. The extension region F**2** extends from the clamping region F**1** to the first elastic piece SP**1**, and the second elastic piece SP**2** is provided with a structural strength weakening part WS corresponding to the extension region F**2**. Specifically, because the extension region F**2** may weaken the structural strength of the first elastic piece SP**1**, the structural strength of the second elastic piece SP**2** may be weakened by the structural strength weakening part WS, to match the structural strength of the first elastic piece SP**1**. The structural strength weakening part WS may be an opening penetrating the second elastic piece SP**2**, which is not limited thereto. In another embodiment, the structural strength weakening part WS may be a plurality of holes penetrating the second elastic piece SP**2** or another material different from a material of the second elastic piece SP**2**, thereby weakening the structural strength of the second elastic piece SP**2**. In this way, in a case the structural strength of the first elastic piece SP**1** and the second elastic piece SP**2** is changed, the first elastic piece SP**1** and the second elastic piece SP**2** can be maintained to push against the sleeve **120** with approximately same thrusts, so that a frame **131***a* can clamp the light uniform column **110** with approximately same forces on two opposite sides connecting the first elastic piece SP**1** and the second elastic piece SP**2**, to prevent the light uniform column **110** from tilting due to uneven thrusts exerted by the first elastic piece SP**1** and the second elastic piece SP**2** on the sleeve **120**.

[0035] Still refer to FIG. **6**. In this embodiment, the extension region F**2**, for example, extends from the clamping region F**1** to a pushing part **1321***a* of the first elastic piece SP**1** and can penetrate the first elastic piece SP**1** in a direction Y. In addition, the structural strength weakening part WS can penetrate a pushing part **1321***b* and a guide part **1322***b* of the second elastic piece SP**2** in the direction Y, and be separated from the clamping region F**1**. However, because features of the structural strength weakening part WS can be changed based on features of the extension region F**2**, the disclosure does not limit this. In addition, the clamping region F**1** in this embodiment, for example, corresponds to the open end **121** of the sleeve **120**, and the light inlet end **111** of the light uniform column **110** can stretch out of the sleeve **120** through the clamping region F**1**. In addition, the extension region F**2** may be corresponding to a side wall **123** of the sleeve **120**, that is, the extension region F**2** can expose a part of the side wall **123** after the end fastening member **130***a* is fastened to the sleeve **120**. Incidentally, an angle A**1** between the pushing part **1321***a* of the first elastic piece SP**1** and the clamping region F**1** may be smaller than an angle A**2** between the pushing part **1321***b* of the second elastic piece SP**2** and the frame **131***a*.

[0036] FIG. **7** is a schematic cross-sectional diagram of a light uniform module according to another embodiment of the disclosure. FIG. **8** is a schematic cross-sectional view of the light uniform module, taken along the line K**3**-K**3** in

FIG. **7**. A structure and advantages of a light uniform module **100***b* in this embodiment are similar to the structure and advantages of the embodiment in FIG. **1**, and only differences are described below. Refer to FIG. **7** and FIG. **8**. For example, the light uniform module **100***b* further includes at least one glue material **140**, and for example, a plurality of glue materials **140** are used in this embodiment. A side wall **123** is provided with, for example, at least one glued through hole H, and for example, a plurality of glued through holes H are used in this embodiment. The glue material **140** is disposed in the glued through hole H to connect at least two of a light uniform column **110**, a sleeve **120**, and an end fastening member **130**. In this way, the glue material **140** can strengthen the fastening of the light uniform column **110**, the sleeve **120**, and the end fastening member **130** to enhance the assembly firmness of the light uniform module **100***b*. In this embodiment, the glue material **140** may be disposed in the glued through hole H after assembly for the light uniform module **100***b* is completed. Further, one part of the glue material **140** may be connected to the light uniform column **110**, a pressing piece P**1**, and a pressing piece P**2**, and the other part of the glue material **140** may be connected to the light uniform column **110** and two elastic pieces **132**. In one embodiment, the glue material **140** may be connected to the light uniform column **110**, the sleeve **120**, and the end fastening member **130**. Incidentally, the glue material **140** in this embodiment includes, for example, an ultraviolet curing adhesive, but the disclosure is not limited thereto.

[0037] FIG. **9** is a schematic block diagram of a projection device according to an embodiment of the disclosure. Refer to FIG. **3** and FIG. **9**. A projection device **200** includes a light source module **210**, a light uniform module **100**, a light valve **220**, and a projection lens **230**. The light source module **210** is configured to provide an illumination beam L**1**. The light uniform module **100** is disposed on a transmission path of the illumination beam L**1**. The light uniform module **100** may be any of the light uniform modules in the above embodiments. The light uniform module **100** is configured to receive the illumination beam L**1**, and the illumination beam L**1** enters a light uniform column **110** through a light inlet surface IS. The illumination beam L**1** is totally reflected in the light uniform column **110** and transmitted to a light outlet surface ES and leaves the light uniform column **110** from the light outlet surface ES. The light valve **220** is disposed on the transmission path of the illumination beam L**1** and is configured to convert the illumination beam L**1** emitted from the light uniform module **100** to an image beam L**2**. The projection lens **230** is disposed on a transmission path of the image beam L**2** and is configured to project the image beam L**2**.

[0038] The light source module **210** may include an excitation light source. In detail, the excitation light source includes, for example, a light emitting diode (LED) or a laser diode (LD), and a quantity of the light-emitting diodes or laser diodes may be one or more. For example, when there is a plurality of LEDs (or LDs), the LEDs (or LDs) may be arranged in a matrix. In an embodiment, the excitation light source includes, for example, a monochromatic light source and a wavelength conversion component may be disposed downstream of an optical path of the excitation light source. In another embodiment, the excitation light source may include a multi-color light source.

[0039] In this embodiment, the light inlet surface IS of the light uniform column **110** may be used for the illumination beam L**1** to enter light guide space G, and is transmitted to the light outlet surface ES in the light guide space G in a total reflection manner, and then leaves the light uniform column **110** from the light outlet surface ES. Specifically, the light inlet surface IS and the light outlet surface ES may include openings that communicate with each other in the light guide space G for the illumination beam L**1** to penetrate. It can be understood that the stop end **122** of the sleeve **120** may be provided with an opening that communicates with the light guide space G so that the illumination beam L**1** can be emitted from the light uniform module **100**. Other features of the light uniform module **100** have been described above, and no redundant detail is to be given herein.

[0040] The light valve **220** includes, for example, a digital micromirror device (DMD), a liquid crystal on silicon (LCoS), or a liquid crystal display panel (LCD), but the disclosure is not limited thereto. In addition, a quantity of the light valves **220** is not limited in the disclosure. For example, the projection device **200** in this embodiment may adopt an architecture of a single-piece liquid crystal display panel or a three-piece liquid crystal display panel, but the disclosure does not limit this.

[0041] The projection lens **230** in this embodiment may include one or more optical lenses, and a diopter brightness of the optical lenses may be the same or different from each other. For example, the optical lens may include various non-planar lenses, such as biconcave lenses, biconvex lenses, concave-convex lenses, convex-concave lenses, plano-convex lenses, and plano-concave lenses, or any combination of the above non-planar lenses. In addition, the projection lens **230** may further include a flat optical lens. A specific structure of the projection lens **230** is not limited in the disclosure.

[0042] Compared with the existing technology, the light uniform module **100** is used for the projection device **200** in this embodiment, so that image quality can be improved, with the product yield and product durability taken into account. It can be understood that, in an embodiment, the projection device **200** may alternatively be a light uniform module **100***a* or **100***b*.

[0043] FIG. **10** is a schematic diagram of a projection device according to another embodiment of the disclosure. A structure and advantages of a projection device **200***a* in this embodiment are similar to the structure and advantages of the embodiment in FIG. **9**, and only the differences are described below. Refer to FIG. **10**. The projection device **200***a* includes, for example, a base **240** and a fastened member **250**. A light uniform module **100** is disposed on the base **240**. The fastened member **250** is fastened on the base **240** and pressed against a sleeve **120** of the light uniform module **100**, to fasten the light uniform module **100** on the base **240**. In detail, the fastened member **250** may be provided with two fastened parts **251**, a body part **252**, and two first pressing parts **253**. The two fastened parts **251** are connected to two opposite sides of the body part **252**. The two first pressing parts **253** are connected to two opposite sides of the body part **252** and are respectively located between the two fastened parts **251**. The two fastened parts **251** are fastened to the base **240**. The body part **252** covers the sleeve **120** of the light uniform module **100**, and the two first pressing parts **253** are pressed against the sleeve **120**. Further, the body part **252** can protect the light uniform module **100**, and the two first pressing parts **253** can prevent the light uniform module **100** from shifting relative to the base **240**. Incidentally, the two fastened parts **251** may be attached to the base **240** through a screw lock, which is not limited thereto in another embodiment.

[0044] The sleeve **120** in this embodiment is provided with, for example, a first pressed wall **124**, and the first pressed wall **124** is located between the light uniform column **110** and the two first pressing parts **253** of the fastened member **250**. The first pressed wall **124** may be provided with a main body **1241** and a protective reinforcing part **1242**. The main body **1241** extends from a stop end **122** of the sleeve **120** to an open end **121**. The protective reinforcing part **1242** is connected to one side of the main body **1241** located at the open end **121**, and the two first pressing parts **253** are pressed against the main body **1241** and/or the protective reinforcing part **1242**. For example, in this embodiment, one of the first pressing parts **253** is pressed against the main body **1241**, and the other of the first pressing parts **253** is pressed against the main body **1241** and the protective reinforcing part **1242**. In this way, the protective reinforcing part **1242** can stop the first pressing part **253** at the open end **121** of the sleeve **120**, preventing the first pressing part **253** from moving from the main body **1241** to the light uniform column **110** and scratching the light uniform column **110**. Incidentally, the protective reinforcing part **1242** is for example, plate-shaped, and the protective reinforcing part **1242** can be bent relative to the main body **1241** in a direction away from the light uniform column **110**, to exactly stop the first pressing part **253** on the main body **1241**. The protective reinforcing part **1242** is not limited thereto. In another embodiment, the protective reinforcing part **1242** may protrude from the open end **121** of the sleeve **120** to stop the first pressing part **253** on the main body **1241**.

[0045] In addition, the fastened member **250** may further be provided with two second pressing parts **254**. The body part **252** of the fastened member **250** is provided with a first plate body **2521** and a second plate body **2522** connected to each other. The two first pressing parts **253** connect two opposite sides of the first plate body **2521**, and the two second pressing parts **254** connect two opposite sides of the second plate body **2522**. The sleeve **120** has a second pressed wall **125** adjacent to the first pressed wall **124**. The first plate body **2521** corresponds to the first pressed wall **124**, and the two first pressing parts **253** are pressed against the first pressed wall **124**. The second plate body **2522** is corresponding to the second pressed wall **125**, and the two second pressing parts **254** are pressed against the second pressed wall **125**. In this way, the two first pressing parts **253** and the two second pressing parts **254** can simultaneously press the light uniform module **100** against the base **240** to strengthen the firmness that the light uniform module **100** is fastened to the base **240**. Incidentally, the second pressed wall **125** is provided with, for example, the protective reinforcing part **1251**, and the protective reinforcing part **1251** can protrude from the open end **121** of the sleeve **120**, to prevent the second pressing part from scratching the light uniform column **110**. It should be noted that the protective reinforcing part **1251** is, for example, the protruding part C in FIG. **1**, the first pressed wall **124** may be the wall part **1232** in FIG. **1**, and the second pressed wall **125** may be the wall part **1231** in FIG. **1**. In another embodiment, the protective reinforcing part **1251** may alternatively be the protective reinforcing part **1242** in a plate shape, which is not limited thereto.

[0046] In summary, for the light uniform module in the disclosure, the sleeve is used to fasten one side of the light uniform column close to the light outlet end, and the end fastening member is used to fasten the light uniform column on one side close to the light inlet end. In this way, a structure used to interfere with the light uniform column near the open end may be omitted for the sleeve, so that a thicker side (i.e., the side close to the light outlet end) of the light uniform column is not damaged by excessive interference of the structure while extending into the sleeve from the open end. In addition, the end fastening member uses the clamping opening of the frame to interfere with a thinner side (i.e., the side close to the light inlet end) of the light uniform column. Therefore, when the end fastening member is sleeved from the light inlet end to the light uniform column, the frame does not touch the light uniform column before interfering with the light uniform column, and the frame can achieve the effect of fastening the light uniform column by slight interference, thereby reducing the damage of the end fastening member to the light uniform column. In addition, because the end fastening member and the sleeve both have a function of fastening the light uniform column, an interference amount of the sleeve to the light uniform column can be further reduced. In this case, the light uniform column, the sleeve, and the end fastening member can still be firmly fastened to each other. Based on the above structure, the light uniform module in the disclosure has an advantage of being easily assembled, with an assembly yield and assembly firmness taken into account. The light uniform module is used for the projection device in the disclosure, so that image quality can be improved, with the product yield and product durability taken into account.

[0047] The foregoing description of the preferred embodiment of the disclosure has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure to the precise form or to exemplary embodiments disclosed. Accordingly, the foregoing description should be regarded as illustrative rather than restrictive. Obviously, many modifications and variations will be apparent to practitioners skilled in this art. The embodiments are chosen and described in order to best explain the principles of the disclosure and its best mode practical application, thereby to enable persons skilled in the art to understand the disclosure for various embodiments and with various modifications as are suited to the particular use or implementation contemplated. It is intended that the scope of the disclosure be defined by the claims appended hereto and their equivalents in which all terms are meant in their broadest reasonable sense unless otherwise indicated. Therefore, the term "the disclosure" is not necessary limited the claim scope to a specific embodiment, and the reference to particularly preferred exemplary embodiments of the disclosure does not imply a limitation on the disclosure, and no such limitation is to be inferred. The disclosure is limited only by the spirit and scope of the appended claims. Moreover, these claims may refer to use "first", "second", etc. following with noun or element. Such terms should be understood as a nomenclature and should not be construed as giving the limitation on the number of the elements modified by such nomenclature unless specific number has been given. The abstract of the disclosure is provided to comply with the rules requiring an abstract, which will allow a searcher to quickly ascertain the subject matter of the technical disclosure of any patent issued from this disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. Any advantages and benefits described may not apply to all embodiments of the disclosure. It should be appreciated that variations may be made in the embodiments described by persons skilled in the art without departing from the scope of the disclosure as defined by the following claims. Moreover, no element and component in the disclosure is intended to be dedicated to the public regardless of whether the element or component is explicitly recited in the following claims.

What is claimed is:

1. A light uniform module, comprising a light uniform column, a sleeve, and an end fastening member, wherein:

the light uniform column is provided with a light inlet end and a light outlet end opposite to each other, the light inlet end is provided with a light inlet surface, the light outlet end is provided with a light outlet surface, and an area of the light inlet surface is smaller than an area of the light outlet surface;

the sleeve is provided with an open end and a stop end opposite to each other, the sleeve is sleeved outside the light uniform column to accommodate the light uniform column, and the light outlet end of the light uniform column faces the stop end; and

the end fastening member is disposed at the open end of the sleeve to fasten relative positions of the light uniform column and the sleeve, and the end fastening member is provided with a frame to form a clamping opening, wherein the light inlet end of the light uniform column penetrates the clamping opening and stretches out of the sleeve from the open end of the sleeve.

2. The light uniform module according to claim **1**, wherein the sleeve further has a side wall, the side wall is located between the open end and the stop end and surrounds the light uniform column, the end fastening member is further provided with two elastic pieces, and the two elastic pieces are respectively connected to two opposite sides of the frame, wherein when the end fastening member is fastened to the open end of the sleeve, the two elastic pieces are located in the sleeve and separately push against the side wall in directions away from each other.

3. The light uniform module according to claim **2**, wherein each of the two elastic pieces is provided with a pushing part and a guide part, the pushing part is connected to the frame, the frame and the guide part are located on two opposite sides of the pushing part, the guide part is connected to the pushing part, and the guide part is bent toward the other guide part relative to the connected pushing part.

4. The light uniform module according to claim **3**, wherein an angle between the guide part and the pushing part of each of the two elastic pieces is an obtuse angle.

5. The light uniform module according to claim **2**, wherein the two elastic pieces comprise a first elastic piece and a second elastic piece, the clamping opening is provided with a clamping region and an extension region communicated with each other, the light uniform column penetrates the clamping region, the extension region extends from the clamping region to the first elastic piece, and the second elastic piece is provided with a structural strength weakening part corresponding to the extension region.

6. The light uniform module according to claim **1**, wherein the end fastening member is further provided with at least one limiting structure, the at least one limiting structure is connected to the frame, and the at least one limiting structure is configured to lean against the open end from a supporting surface of the open end of the sleeve.

7. The light uniform module according to claim **6**, wherein the open end of the sleeve is provided with an open part and a protruding part, the light uniform column stretches out from the open part, the protruding part protrudes from the open part in a direction away from the stop end, the supporting surface is located at the protruding part, the at least one limiting structure is provided with an extension section and a stop section, the extension section is connected to the frame, the frame and the stop section are located on two opposite sides of the extension section, the stop section is connected to the extension section and bent relative to the extension section, and the stop section is configured to lean against the protruding part from the supporting surface of the sleeve.

8. The light uniform module according to claim **1**, wherein the sleeve further has a side wall, the side wall is located between the open end and the stop end and surrounds the light uniform column, the side wall is provided with two wall parts adjacent to each other, each of the two wall parts is provided with a wall body and a pressing piece connected to each other, and the pressing piece is configured to press the light uniform column to be close to an outer surface of the light outlet end.

9. The light uniform module according to claim **8**, wherein the wall body and the pressing piece are integrally formed, or the pressing piece is a piece additionally connected to the wall body.

10. The light uniform module according to claim **1**, further comprising at least one glue material, wherein the sleeve further has a side wall, the side wall is located between the open end and the stop end and surrounds the light uniform column, the side wall is provided with at least one glued through hole, and the at least one glue material is disposed in the at least one glued through hole to connect at least two of the light uniform column, the sleeve, and the end fastening member.

11. The light uniform module according to claim **1**, wherein an opening area of the clamping opening is between an area of the light inlet surface of the light uniform column and an area of the light outlet surface.

12. A projection device, comprising a light source module, a light uniform module, a light valve, and a projection lens, wherein:

the light source module is configured to provide an illumination beam;

the light uniform module is disposed on a transmission path of the illumination beam, the light uniform module comprises a light uniform column, a sleeve, and an end fastening member, the light uniform column is configured to receive the illumination beam, the light uniform column is provided with a light inlet end and a light outlet end opposite to each other, the light inlet end is provided with a light inlet surface, the light outlet end is provided with a light outlet surface, the illumination beam enters the light uniform column from the light inlet surface, the illumination beam is totally reflected in the light uniform column and transmitted to the light outlet surface and leaves the light uniform column from the light outlet surface, an area of the light inlet surface of the light uniform column is smaller than an area of the light outlet surface, the sleeve is provided with an open end and a stop end opposite to each other, the sleeve is sleeved outside the light uniform column to accommodate the light uniform column, the light outlet end of the light uniform column faces the stop end, the end fastening member is disposed at the open end of the sleeve to fasten relative positions of the light uniform column and the sleeve, the end fastening member is provided with a frame to form a clamping opening, and the light inlet end of the light uniform column penetrates the clamping opening and stretches out of the sleeve from the open end of the sleeve;

the light valve is disposed on the transmission path of the illumination beam and is configured to convert the illumination beam emitted from the light uniform module to an image beam; and

the projection lens is disposed on a transmission path of the image beam and is configured to project the image beam.

13. The projection device according to claim **12**, further comprising a base and a fastened member, wherein the light uniform module is disposed on the base, and the fastened member is fastened to the base and pressed against the sleeve of the light uniform module.

14. The projection device according to claim **13**, wherein the fastened member is provided with two fastened parts, a body part, and two first pressing parts, the two fastened parts are respectively connected to two opposite sides of the body part, the two first pressing parts are respectively connected to the two opposite sides of the body part and are separately located between the two fastened parts, the two fastened parts are fastened to the base, the body part covers the sleeve of the light uniform module, and the two first pressing parts are pressed against the sleeve.

15. The projection device according to claim **14**, wherein the sleeve is provided with a first pressed wall, the first pressed wall is located between the light uniform column and the two first pressing parts of the fastened member, the first pressed wall is provided with a main body and a protective reinforcing part, the main body extends from the stop end of the sleeve to the open end, the protective reinforcing part is connected to one side of the main body located at the open end, and the two first pressing parts are pressed against the main body and/or the protective reinforcing part.

16. The projection device according to claim **15**, wherein the protective reinforcing part is plate-shaped, and the protective reinforcing part protrudes from the open end or is bent relative to the main body in a direction away from the light uniform column.

17. The projection device according to claim **14**, wherein the fastened member is further provided with two second pressing parts, the body part of the fastened member is provided with a first plate body and a second plate body connected to each other, the two first pressing parts are connected to two opposite sides of the first plate body, the two second pressing parts are connected to two opposite sides of the second plate body, the sleeve is provided with a first pressed wall and a second pressed wall adjacent to each other, the first plate body is corresponding to the first pressed wall, the two first pressing parts are pressed against the first pressed wall, the second plate body is corresponding to the second pressed wall, and the two second pressing parts are pressed against the second pressed wall.

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